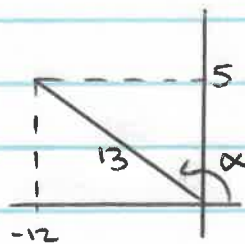


Grade 12
Cycle Test

23 April

Question 1

$$\cos \alpha = -\frac{12}{13}$$



✓ Quad
✓ $y=5$

$$y^2 = 13^2 - 12^2$$

$$y = \pm 5 \therefore y = 5$$

$$\sin(\alpha - \beta)$$

$$= \sin \alpha \cos \beta - \cos \alpha \sin \beta \quad \checkmark$$

$$= \frac{5}{13} \left(-\frac{6}{10} \right) - \left(-\frac{12}{13} \right) \left(-\frac{8}{10} \right) \quad \checkmark$$

✓ $r=10$

$$= -\frac{63}{65} \quad \checkmark$$

(6)

Question 2

$$2.1. \sin^2 15^\circ - \cos^2 15^\circ$$

$$= -(\cos^2 15^\circ - \sin^2 15^\circ) \checkmark$$

$$= -\cos 30^\circ \checkmark$$

$$= -\frac{\sqrt{3}}{2} \checkmark$$

(3)

$$2. \sin 140^\circ \cos 340^\circ - \cos 40^\circ \sin(-20^\circ)$$

$$= \sin 40^\circ \cos 20^\circ + \cos 40^\circ \sin 20^\circ \checkmark$$

$$= \sin(40^\circ + 20^\circ)$$

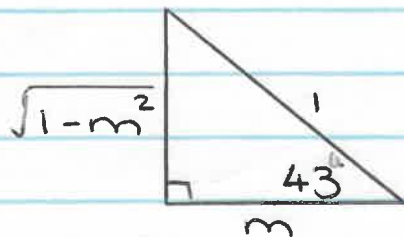
$$= \sin 60^\circ \checkmark$$

$$= \frac{\sqrt{3}}{2} \checkmark$$

(5)

Question 3

3.1. $\cos 43^\circ = m$



$\tan 137^\circ$

$= -\tan 43^\circ$ ✓

$= -\frac{\sqrt{1-m^2}}{m}$ ✓ ✓

(3)

3.2. $\sin 86^\circ$

$= 2\sin 43^\circ \cos 43^\circ$ ✓

$= 2m\sqrt{1-m^2}$ ✓

(2)

3.3. $\cos 17^\circ$

$= \cos (60^\circ - 43^\circ)$ ✓

$= \cos 60^\circ \cos 43^\circ + \sin 60^\circ \sin 43^\circ$ ✓

$= \frac{1}{2}m + \frac{\sqrt{3}}{2}\sqrt{1-m^2}$ ✓

(3)

Question 4

$$+1. \frac{\cos(20^\circ - x) \cdot \cos x - \sin(20^\circ - x) \cdot \sin x}{\sin 215^\circ \cdot \cos 325^\circ}$$

$$= \frac{\cos(20^\circ - x + x)}{-\sin 35^\circ \cdot \cos 35^\circ}$$

$$= \frac{\cos 20^\circ}{-\sin 35^\circ \cdot \cos 35^\circ} \times \frac{2}{2}$$

$$= \frac{2 \cos 20^\circ}{-\sin 70^\circ}$$

$$= \frac{2 \cos 20^\circ}{-\cos 20^\circ} \quad \text{or} \quad \frac{2 \sin 70^\circ}{-\sin 70^\circ}$$

$$= -2$$

(7)

$$+2. 2\cos^2 22,5^\circ - \sin^2 48^\circ - \sin^2 42^\circ$$

$$= 2\cos^2 22,5^\circ - \sin^2 48^\circ - \cos^2 48^\circ \quad \text{co-function}$$

$$= 2\cos^2 22,5^\circ - (\sin^2 48^\circ + \cos^2 48^\circ)$$

$$= 2\cos^2 22,5^\circ - 1$$

$$= \cos 45^\circ$$

$$= \frac{\sqrt{2}}{2}$$

(4)

Question 5

$$5.1. \quad 2\sin\theta\cos\theta = 2 - 4\sin^2\theta$$

$$\sin 2\theta = 2(1 - 2\sin^2\theta)$$

$$\sin 2\theta = 2\cos 2\theta$$

$$\tan 2\theta = 2$$

$$\text{ref } \angle = 63,43^\circ$$

$$2\theta = 63,43^\circ + k \cdot 180^\circ$$

$$\theta = 31,72^\circ + k \cdot 90^\circ; \quad k \in \mathbb{Z}$$

(6)

Question 6

$$6.1. \frac{1 - \tan x}{1 + \tan x} = \frac{\cos 2x}{1 + \sin 2x}$$

$$\text{LHS} = \frac{1 - \frac{\sin x}{\cos x}}{1 + \frac{\sin x}{\cos x}} \quad \checkmark \tan x = \frac{\sin x}{\cos x}$$

$$= \frac{\cos x - \sin x}{\cos x} \div \frac{\cos x + \sin x}{\cos x}$$

$$= \frac{\cos x - \sin x}{\cos x} \times \frac{\cos x}{\cos x + \sin x} \quad \checkmark m$$

$$= \frac{\cos x - \sin x}{\cos x + \sin x} \quad \checkmark$$

$$\text{RHS} = \frac{\cos^2 x - \sin^2 x}{1 + 2\sin x \cos x} \quad \checkmark$$

$$= \frac{\cos^2 x - \sin^2 x}{\cos^2 x + 2\sin x \cos x + \sin^2 x} \quad \checkmark 1 = \sin^2 + \cos^2$$

$$= \frac{(\cos x + \sin x)(\cos x - \sin x)}{(\cos x + \sin x)(\cos x + \sin x)} \quad \checkmark \text{fact}$$

$$= \frac{\cos x - \sin x}{\cos x + \sin x}$$

$$\therefore \text{LHS} = \text{RHS}$$

(7)

$$o.2. \quad x = 90 + k \cdot 180^\circ ; k \in \mathbb{Z} \quad \checkmark$$

or

$$1 + \tan x = 0$$

$$\tan x = -1$$

or

$$1 + \sin 2x = 0$$

$$\sin 2x = -1$$

$$x = 135 + k \cdot 180$$

$$k \in \mathbb{Z}$$

$$2x = 270 + k \cdot 360$$

$$x = 135 + k \cdot 180^\circ \quad \checkmark$$

$$k \in \mathbb{Z}.$$

(4)
